IT1244
Application of supervised training models to a utility fraud dataset

Group 10: Arkan Alexei Andrei (A0290262J), Jiang Ruiqian (A0282309A), Kang Hao Yu (A0233719B), Tey Zi Pin (A0252547B)

Introduction

Fraud detection in electricity and gas consumption is a critical concern for utility companies worldwide. In general, illegal connections in the energy sector can be categorized into two types: direct connections without an authorized supply contract and activities tampering with the supply connection, potentially affecting the meter and associated installation. Regardless of the type, the consequence remains the same: illegal connection to the grid without adhering to safety regulations, maintenance standard as well as the terms and conditions for use. Illegal connections are not only fraudulent activities, but also pose significant risks to both human safety and the whole electrical system. Tampering with electrical installations can result in serious health hazards such as electric shock and fatalities, while disrupting the reliability of power supply. Furthermore, the economic burden of electricity fraud extends to consumers through increased utility cost. Recognizing the severity of the consequences, it is essential for electricity companies to detect and prevent such fraudulent behaviors, safeguarding both individuals and the electricity supply infrastructure.

Traditionally, fraud detection in energy consumption has relied on rule-based systems (Otuoze et al., 2022). These systems employ predefined rules to identify potential instances of fraud based on certain patterns or conditions. Despite being relatively easy to set up and capable of quickly detecting clear-cut cases of fraud, rule-based system has inherent limitations. They often struggle to adapt to new or evolving fraud patterns, making them less effective against sophisticated attacks.

This project aims to analyze consumption patterns and identify anomalies through machine learning (ML) and artificial intelligence (AI) techniques. Supervised machine learning algorithms including Random Forest Classifier (RF) and K-Nearest Neighbors (KNN) are used. These algorithms are suitable for classification tasks where we aim to predict whether a given instance belongs to a fraudulent or non-fraudulent class based on labeled training data. Random Forest is chosen for its ability to handle complex datasets with high dimensionality and nonlinear relationships between features. KNN, on the other hand, is a simple yet effective algorithm for pattern recognition and classification tasks, making it suitable for our dataset. While some researchers have investigated using sophisticated, highly complex solutions including Artificial Neural Networks (ANN), Deep Artificial Neural Networks (DANN) and AdaBoost on electricity fraud (Bohani et al., 2021), we wanted to explore the effectiveness of utilising computationally simpler methodologies to resolve the problem.

Dataset

The dataset contains two CSV files: ‘client.csv’ and ‘invoice.csv’. There are over 21000 people, with each of them having 3-50 invoices. The ‘client.csv’ file contains information about clients, including district, category, region, date and target label indicating fraud or non-fraud. The ‘invoice.csv’ file contains 15 details of the invoice, such as tariff type, counter readings, consumption levels, and date of the invoice. There are some issues with the dataset and several steps were performed to improve the quality of the data, address imbalance issues, and select relevant features to build effective predictive models.

Imbalanced dataset

One major issue with the dataset is the imbalance between fraudulent and non-fraudulent cases. There are very few fraud instances, which can lead to biased model performance. To address this issue, techniques such as using alternative evaluation metrics (e.g., precision, recall, F1-score), undersampling, oversampling (e.g., SMOTE), and ensemble methods are employed.

Data preprocessing

- The client and invoice datasets were merged based on the client ID to create a unified dataset with 21 attributes.
- Variable types were changed to the correct format, including date and counter number.

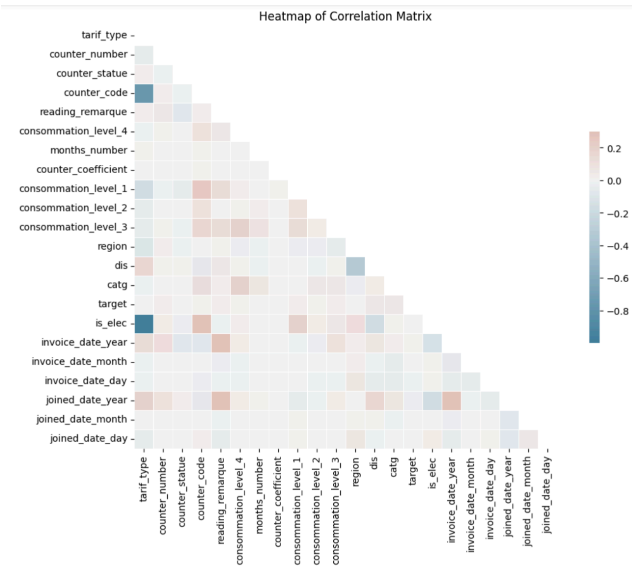
- Categorical variable was encoded. Since the counter_type column only has two possible values, it is converted to is_elec.
- Unnecessary ID columns were removed.
- Date components were separated into individual columns for analysis.

Feature engineering

- Delta Index: Computing the difference between old and new index values to represent changes in consumption levels.

Feature selection

The pairwise correlation matrix was visualized to identify highly correlated features. Features with a correlation coefficient above a certain threshold (e.g., 0.1) with the target variable were selected.



However, there were no columns that are highly correlated to the target column. We were unable to derive any meaningful differences between using any particular feature to predict the target variable. Thus, we opted to use all variables in running our models.

Handling Missing Values

Missing values in the dataset were checked. Since there are no missing values, no imputation or handling is required.

Handling Duplicated Values

Duplicated rows in the dataset were identified and removed to ensure data integrity and avoid bias in the analysis.

Methods

As mentioned above, the current problem can be considered a classificatory one, where the model is required to identify new fraudulent clients based on the details of their transactions. Classificatory problems can be approached from multiple machine learning perspectives, including both supervised and unsupervised learning. While supervised learning has the advantages of having higher predictive power and more easily interpretable results, a drawback is the need for labeled data, which might not be readily available for all problems. This can be seen from the current literature on electricity fraud where obtaining labelled data remains difficult and unsupervised ML techniques have even been used to aid in flagging out potential fraud cases (Oprea & Bâra, 2021). However, in our case, the dataset provided labels for which clients are fraudulent. Considering these factors, we decided to approach the problem with supervised machine learning algorithms. These algorithms are able to generate predictive models based on identifying the relationships and patterns of the input data and labels. In the case of our problem, the model should be able to predict which clients are fraudulent based on the history of their transactions.

We utilized two different supervised machine learning algorithms to model the data. These were the K-Nearest Neighbours (KNN) and Random Forest Classifier (RF) models. We also split the dataset into training and testing sets. The models were trained based on an 80/20 training-testing random split. We performed a stratified split to ensure that the distribution of fraudulent and non-fraudulent cases is similar in both sets. The split was chosen to ensure an optimal balance between providing sufficient data for model learning and robustly evaluating predictive performance, while also ensuring computational efficiency and mitigating bias-variance trade-offs. Results obtained from the testing stage were evaluated and compared using 5 metrics: Accuracy, Precision, Recall, F1 Score and ROC AUC score.

K-Nearest Neighbours (KNN)

We decided to include KNN as a baseline to compare the results obtained from RF. KNN is a lazy-learning algorithm which stores the entire dataset and compares new datapoints to ones already in the dataset. Our KNN implementation includes hyperparameter tuning using GridSearchCV which allows us to select the optimal K value for the KNN model.

Data Standardization

Since the dataset contains features of various scales, standardization is performed to bring all features to a similar scale, which helps improve the performance of KNN.

Handling Imbalanced Classes Using SMOTE

Synthetic Minority Over-sampling Technique (SMOTE) is applied to address the issue of class imbalance by over-sampling the minority class (fraudulent cases) to achieve a balanced distribution of classes in the training data.

Random Forest Classifier (RF)

RF is a supervised machine learning algorithm which generates multiple decision trees and considers the combined output of these trees to reach a result. Instead of using the training dataset as it is, the RF algorithm generates random subsets of data based on the training dataset. These subsets each have their own unique decision tree based on random features from the dataset. The results from each decision tree are considered and the class predicted by majority of the decision trees is taken as the final prediction. The RF can be considered an improved decision tree as it is known to reduce overfitting especially in the cases of large datasets with high dimensionality. SMOTE was also used to balance the dataset before running the RF algorithm.

Results and Discussion

Results

We implemented the two aforementioned models. We performed a 3-fold cross validation for both models.

Table 1. Model Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
KNN	0.897	0.830	0.998	0.906	0.969
RF1	0.953	0.593	0.858	0.702	0.909
RF2	0.953	0.594	0.857	0.701	0.908
RF3	0.953	0.593	0.863	0.703	0.911
RF\[1]	0.953	0.593	0.859	0.702	0.909

[1]: the average of the 3 rounds of RF was taken into consideration for evaluation

From Table 1, it can be seen that there were notable differences in performance for both models based on the 5-metrics we have set. We will now evaluate both models based on the individual matrices.

$$Precision = \frac{TP}{TP+FP}$$

$$Recall = \frac{TP}{TP+FN}$$

$$F1\,Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Figure 1: Formulas for Precision, Recall, F1 Score

Accuracy

Both models performed well based on the accuracy metric. Accuracy is calculated as a ratio of correct predictions to all predictions. Both KNN and RF demonstrate respectable accuracy (KNN = 0.897, RF = 0.953), which indicates that both models were able to frequently predict the correct outcome. However, it is important to note that the accuracy matric is highly biased for imbalanced datasets.

Precision

KNN performed significantly better than RF for the precision metric (KNN = 0.830, RF = 0.593). Precision is measured as the ratio of true positives to the total number of predicted positives. It measures how often a model correctly predicts a positive class. The low precision value for RF suggests that our RF model predicted a high number of false positives in the test dataset. Despite the use of SMOTE to balance the dataset, the RF algorithm was still unable to accurately generalize which resulted in possible overfitting when evaluated with the test dataset.

Recall

Of particular note is the extremely high recall value of the KNN model (KNN = 0.998, RF = 0.859). Recall is measured as the ratio of true positives to the total number of positive instances. In essence, it measures how often the model is able to find an instance of the target class in the dataset. Our high recall value obtained demonstrates that KNN was able to easily find fraudulent clients, although the low false negative rate could also be attributed to the low number of fraudulent clients.

F1-Score

The KNN model returned a higher F1-Score compared to the RF (KNN = 0.906, RF = 0.702). The F1-Score combines precision and recall scores to provide a more balanced measure of a model's performance. The lower F1-Score for the RF as a result of its low precision score suggests that our RF model is skewed towards returning a high number of false positives compared to false negatives. In other words, the RF is more likely to incorrectly classify a client as committing utility fraud when he is not.

ROC-AUC Score

The ROC-AUC score is obtained by mapping the true positive rate against the false positive rate and computing the area under the curve to determine the discriminating ability of the model. The high ROC-AUC scores for both models (KNN = 0.969, RF = 0.909) suggests that both models were able to reliably classify true positive and true negative cases.

Limitations

Metrics

In the context of fraud detection in electricity consumption, accuracy alone is not adequate for evaluating model performance. This is primarily because fraud detection datasets are often highly imbalanced, meaning that the number of non-fraudulent instances (negative class) significantly outweighs the number of fraudulent instances (positive class). Hence, accuracy can be misleading as it may appear high even if the model simply predicts the majority class of non-fraud for all instances. Instead, metrics that focus on the model's ability to correctly identify instances of the minority class of fraudulent transactions are more appropriate. The cost associated with false negatives is often higher than the cost of false positives. For instance, missing actual fraudulent transactions (false negatives) could result in significant revenue losses, safety risks and decreased reliability of energy supply systems, while falsely flagging legitimate transactions as fraud (false positives) only lead to unnecessary investigations. Hence, recall is prioritized over precision as a high recall ensures the model can identify as many actual fraudulent cases as possible, reducing the possibilities of overlooking fraudulent activities.

In addition to recall and precision, F1 score and ROC-AUC are used. F1-score provides a balance between precision and recall, while ROC-AUC assesses the model's discrimination ability across various classification thresholds.

Algorithm Employed

Both KNN and RF are non-parametric ML algorithms which faced similar issues during training. Due to the high number of parameters factored in, both models took a relatively long time to train and cross-validate for our dataset (KNN = 1973s, RF = 571s). This calls into question the efficiency of using such models especially f or large datasets.

Extension/future improvement

The reliance on labelled datasets might not be feasible all the time for the case of utilities fraud, as each dataset comes with inherent bias of the consumption patterns of the local population. Labelling of data for utilities fraud is likely to take a long time due to the need for proper investigation after flagging suspicious activity. As such, a possible alternative to supervised ML algorithms would be to use deep learning techniques. Specifically, recurrent neural networks (RNN) and Convolutional neural networks (CNN) can be explored.

RNN

RNN is designed for sequence problems. In fraud detection, RNN can model the sequential nature of transactions or events associated with electricity consumption. For example, it can analyze patterns of usage over time to detect anomalies or deviations from normal behavior. In addition, RNN can automatically extract relevant features from sequential data without the need for manual feature engineering. This is advantageous in fraud detection tasks where the characteristics of fraudulent behavior may not be immediately apparent.

CNN

CNNs can learn translation-invariant patterns, and thus able to leverage image processing techniques to detect fraudulent activities. For example, CNN can identify irregularities in meter readings or physical installations captured in images, which may indicate tampering or unauthorized connections. Furthermore, CNN employ hierarchical feature learning, where lower layers detect simple local patterns, and higher layers combine these patterns to detect more complex features. This approach enables CNN to learn representations of fraudulent patterns at different levels of abstraction, enhancing its ability to detect more complicated anomalies.

Conclusion

Given a good, labelled dataset, we have shown that basic machine learning models such as KNN and RF are able to achieve high accuracy in identifying potential cases of utilities fraud. The KNN model returned better scores for precision, recall, f1-score, and roc-auc score, achieving a better overall performance compared to the RF model. In part it was because we were unable to do data engineering appropriately such that it complements the RF model. In fact, the research done by Rubaidi, Ammar, and Aouicha (2022) with a similar dataset showed that they achieved the best result using the Rf model across all metrics. Unfortunately, we were not able to replicate their success, as we could not find the right data engineering techniques used in time for submission of this paper. Nevertheless, for large datasets, these models are not the most efficient computationally as they require significant computational resources and time for training and prediction. Gradient Boosting Machine (GBM) algorithms are a viable alternative due to their ability to construct ensemble models sequentially. Alternatively, more dimensionality reduction methods and analysis such as PCA could be conducted on the dataset to determine more relevant factors to feed into the model.

Appendix

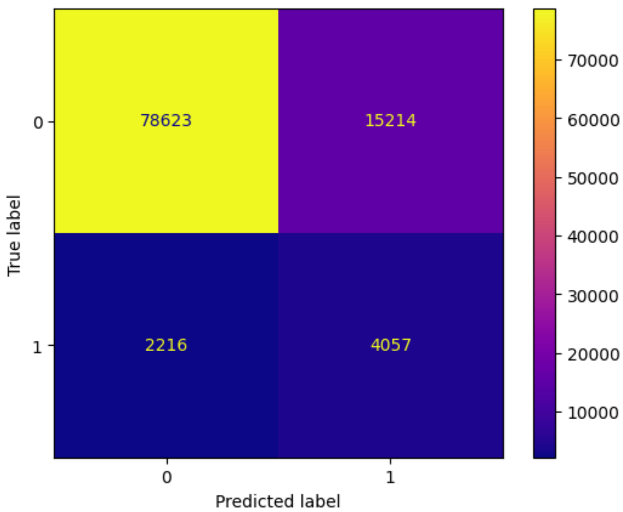


Figure 2: Confusion Matrix for KNN

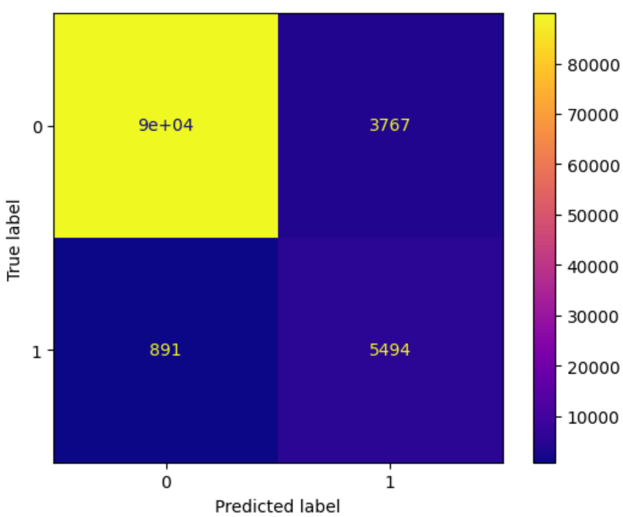


Figure 3: Confusion Matrix for RF

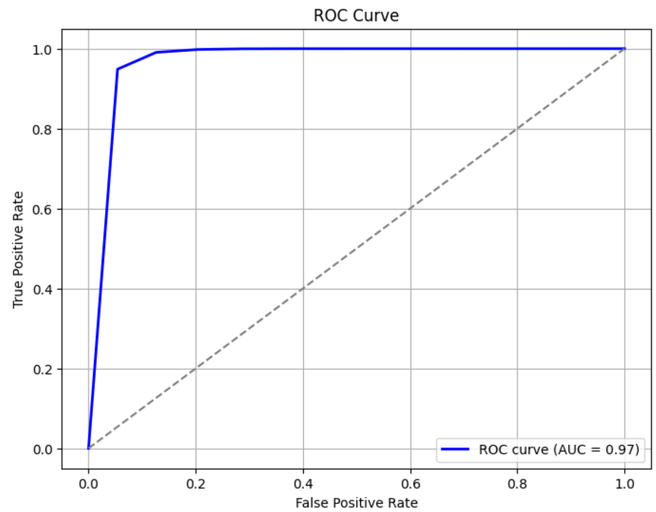


Figure 4: ROC Curve for KNN

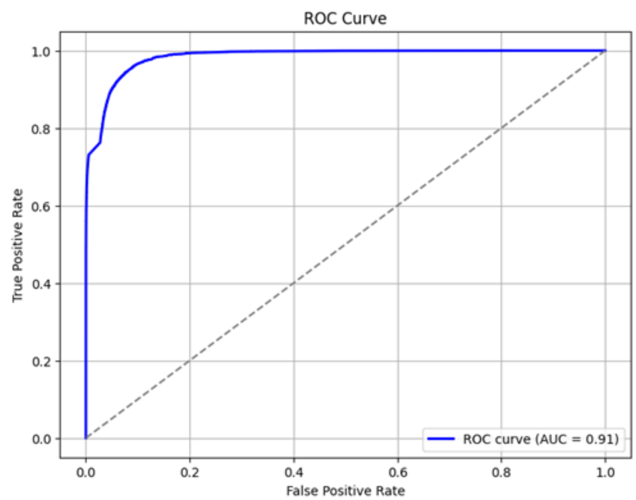


Figure 5: ROC Curve for RF

References

Bohani, F. A., Suliman, A., Saripuddin, M., Sameon, S. S., Md Salleh, N. S., & Nazeri, S. (2021). A comprehensive analysis of supervised learning techniques for electricity theft detection. Journal of Electrical and Computer Engineering, 2021, 1–10. https://doi.org/10.1155/2021/9136206

Brownlee, J. (2020). *SMOTE for Imbalanced Classification with Python.* Machine Learning Mastery. https://machinelearningmastery.com/smote-oversampling-for-imbalanced-classification/.

Endesa (n.d.) *Electricity fraud: what to do*. https://www.endesa.com/en/the-e-face/electrical-grid/electricity-fraud-what-to-do.

Oprea, S.-V., & Bâra, A. (2021). Machine learning classification algorithms and anomaly detection in conventional meters and Tunisian electricity consumption large datasets. Computers & Electrical Engineering, 94, 107329. https://doi.org/10.1016/j.compeleceng.2021.107329

Otuoze, A.O., Mustafa, M.W., Abioye, A.E. (2022) A rule-based model for electricity theft prevention in advanced metering infrastructure. *Journal of Electrical Systems and Inf Technol* **9**, 2. https://doi.org/10.1186/s43067-022-00042-5

Rubaidi, Z. S., Ammar, B. B., & Aouicha, M. B. (2022). Fraud detection using large-scale imbalance dataset. International Journal on Artificial Intelligence Tools, 31(08). https://doi.org/10.1142/s0218213022500373

Sathya, R. & Abraham, Annamma. (2013). Comparison of Supervised and Unsupervised Learning Algorithms for Pattern Classification. International Journal of Advanced Research in Artificial Intelligence, 2(2). http://dx.doi.org/10.14569/IJARAI.2013.020206

TransUnion (n.d.). *What is the most common fraud detection?* https://www.transunion.com/blog/what-is-the-most-common-fraud-detection